

Thermodynamic Behavior Models & Secure Telemetry Architecture

Working notes for SIH26054 — AI-enabled digital twin for MALE UAV piston engine health monitoring
(Team: The Bootloaders)

1. Thermodynamic Behavior Models

This is the physics half of the physics-informed ML approach. For the Rotax 912/914 (four-stroke spark-ignition, with the 914 adding a turbocharger), it means writing equations that predict how the engine *should* behave thermally and mechanically under given flight conditions.

What it covers

- **Combustion & heat release** — how fuel-air ratio, ignition timing, and RPM translate into cylinder heat generation
- **Heat transfer / cooling** — expected Cylinder Head Temperature (CHT) and Exhaust Gas Temperature (EGT) as a function of RPM, load, and ambient air
- **Turbocharger thermodynamics** (914 only) — boost pressure vs. intercooler effectiveness vs. altitude
- **Oil temperature/pressure behavior** under varying load

Why it matters for the digital twin

The model computes what the sensors *should* read given current flight conditions, then compares that to what the sensors actually read. The gap (the residual) is the fault signal — e.g. a rising CHT residual with no RPM change could indicate a cooling fin issue or blocked airflow, before it becomes a full failure. This directly implements the "physics-informed over pure ML" architecture stance already chosen for the project, and reuses the environmental parameter physics work already documented (Climate_Environmental_Parameters.pdf).

2. Secure Telemetry Architecture

Encryption is the core of it, but the deliverable is really three things bundled together:

- **Encryption** (AES-256 is the standard choice) — protects sensor data in transit from being read if intercepted
- **Authentication** — confirms telemetry claiming to come from the UAV actually did, preventing spoofed or injected data
- **Key management** — how encryption keys are provisioned and rotated

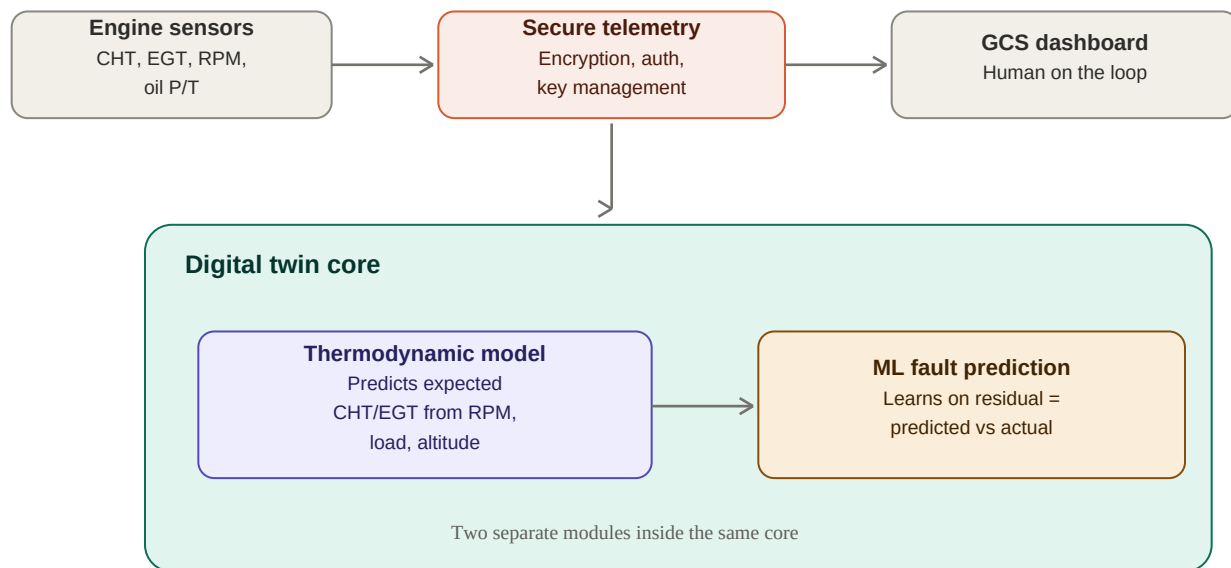
C2 link — how it relates

C2 (Command and Control) is the *uplink* — commands going from the Ground Control Station to the aircraft (throttle, waypoints, flight mode changes). This project is concerned with *telemetry*, the *downlink* — engine sensor data going from the aircraft to the ground or digital twin backend. They are a matched pair (uplink command channel + downlink data channel) but distinct systems.

Recommended scope: secure the engine-health telemetry downlink only, and explicitly state that C2 / flight-control links are out of scope. This keeps the project consistent with the existing "human-on-the-loop, engineering-only OEM involvement" framing, and avoids drifting into actual military C2 systems territory — which is outside the problem statement's remit.

3. Where These Sit in the Architecture

Both are separate architectural components — neither is embedded inside the ML training loop.



- **Thermodynamic model** — computes expected engine behavior; its output (the residual) becomes an *input feature* for the ML model, not part of the ML model itself
- **Secure telemetry** — pure infrastructure between aircraft and ground station; unrelated to the ML pipeline, and effectively a separate deliverable from either the physics or ML work

4. Suggested Next Steps

- Extend existing environmental-parameter physics work into explicit predictive equations for CHT/EGT/oil temp vs. flight phase
- Document the telemetry encryption scheme (AES-256), authentication approach, and key management plan
- Explicitly scope out C2/flight-control links as out of scope, mirroring the existing OEM-involvement and autonomy scoping language
- Cross-check final wording against PLAN.md for consistency with the rest of the national submission